Sleep and Why It Matters

Knowledge document 043 of 100

Introduction to Sleep

Sleep is a natural, recurring state of rest in which consciousness is reduced, muscles relax and the body and brain carry out essential repair and maintenance. People spend about one-third of their lives asleep. Sleep is not a passive shutdown: during sleep the brain consolidates memories, clears waste products, regulates hormones, and the body repairs tissues and strengthens the immune system. Good sleep is as essential to health as food, water and exercise. Lack of sleep is linked to poor concentration, low mood, weak immunity, weight gain, diabetes, heart disease and accidents. World Sleep Day is celebrated every year in March.

How Many Hours of Sleep Do Young Adults Need?

Young adults, aged about 18 to 25 years, generally need 7 to 9 hours of sleep every night, according to the National Sleep Foundation, and the American Academy of Sleep Medicine recommends that adults get at least 7 hours. Teenagers aged 13 to 18 need 8 to 10 hours. School-age children aged 6 to 12 need 9 to 12 hours, and older adults need 7 to 8 hours. Individual needs vary a little, but regularly sleeping less than 6 hours is considered insufficient. Many students get less than they need because of late-night studying, mobile phones and early classes. "Sleep debt" builds up if you miss sleep, and it cannot always be fully repaid by sleeping late on weekends.

The Body Clock

Sleep is controlled by two main systems. The circadian rhythm is the body's internal clock of about 24 hours, driven by a small area of the brain called the suprachiasmatic nucleus, which responds to light and dark. The scientists Jeffrey Hall, Michael Rosbash and Michael Young won the 2017 Nobel Prize in Physiology or Medicine for discovering molecular mechanisms of this clock. The second system is sleep pressure, the buildup of the chemical adenosine during waking hours that makes us feel sleepy. When it gets dark, the brain releases the hormone melatonin, which signals that it is time to sleep. Morning light helps reset the clock.

Stages of Sleep

Sleep happens in cycles of about 90 minutes, and a person goes through four to six cycles in a night. Each cycle has stages. Non-REM sleep has three stages: stage 1 is light sleep when we drift off, stage 2 is a deeper light sleep where heart rate and body temperature fall, and stage 3 is deep sleep, also called slow-wave sleep, which is the most restorative, when the body repairs tissues, builds muscle and releases growth hormone. Deep sleep is more common in the first half of the night.

What Is REM Sleep?

REM sleep stands for Rapid Eye Movement sleep. It is the stage of sleep in which the eyes move rapidly behind closed eyelids, the brain becomes very active, almost as active as when awake, breathing and heart rate become irregular and most vivid dreaming occurs. During REM sleep the body's voluntary muscles are temporarily paralysed, a state called atonia, which prevents people from acting out their dreams. REM sleep was discovered in 1953 by Eugene Aserinsky and Nathaniel Kleitman at the University of Chicago. It makes up about 20 to 25 percent of total sleep in adults, and the first REM period starts about 90 minutes after falling asleep and gets longer through the night, so cutting sleep short reduces REM sleep. REM sleep is important for learning, memory consolidation, creativity and emotional regulation.

How Do Screens Before Bed Affect Sleep?

Using phones, tablets, computers and TVs before bedtime harms sleep in several ways. First, screens emit blue-enriched light, which suppresses the production of melatonin and tricks the brain into thinking it is still daytime, shifting the body clock later. Second, the content, such as social media, games, videos and messages, is mentally stimulating and keeps the mind alert, and notifications interrupt sleep. Third, screen time takes the time that would otherwise be spent sleeping, causing "bedtime procrastination". The result is that people take longer to fall asleep, sleep less deeply and feel tired the next day. Experts advise switching off screens at least 30 to 60 minutes before bed, using night mode or a blue-light filter in the evening, keeping phones out of reach of the bed and charging them outside the bedroom.

Tips for Better Sleep

Go to bed and wake up at the same time every day, including weekends. Keep the bedroom dark, quiet and cool. Avoid caffeine from tea, coffee, cola and energy drinks in the afternoon and evening, since caffeine lasts for 6 hours or more. Avoid heavy meals, alcohol and nicotine near bedtime. Get sunlight in the morning and exercise regularly, but not very close to bedtime. Use the bed only for sleep. A short afternoon nap of 20 to 30 minutes can refresh the mind, but long late naps can disturb night sleep. Relaxing routines such as reading a book, listening to calming music or gentle yoga help.

Sleep Disorders

Common sleep problems include insomnia, which is difficulty falling or staying asleep; sleep apnoea, in which breathing repeatedly stops during sleep, often with loud snoring; restless legs syndrome; and narcolepsy. If sleep problems last for several weeks, a doctor should be consulted.

Sleep in India

Surveys suggest that many Indians sleep later and less than recommended, particularly in cities and among young people using smartphones at night. Traditional Indian systems of medicine, such as Ayurveda, regard nidra (sleep) as one of the three pillars of health along with food and regulated lifestyle. Techniques such as yoga nidra and pranayama are used to improve sleep, and Chennai's hospitals have sleep clinics for diagnosis.

Challenges and Future

Modern lifestyles with shift work, long commutes and screens challenge healthy sleep. Wearable sleep trackers, smart lighting and research on sleep and memory are helping people sleep better.

Quick Facts

- Young adults need about 7 to 9 hours of sleep a night.
- REM sleep is Rapid Eye Movement sleep, with vivid dreams and high brain activity; it makes up 20 to 25 percent of sleep.
- Screens before bed suppress melatonin with blue light and keep the mind stimulated, delaying sleep.
- A sleep cycle lasts about 90 minutes.
- REM sleep was discovered in 1953 by Aserinsky and Kleitman.